Improved accuracy of on-line heavy water measurement using infrared spectroscopy by investigation of signal-to-noise ratio and temperature influences

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Experimental protocols for the on-line measurement of heavy water concentration in nuclear power plants have been established, and important parameters, such as the temperature and signal-to-noise ratio, which govern the accuracy of measurement, have been studied. The temperature of a sample should be controlled tightly because the temperature variation introduces non-linear baseline variations and leads to an increase of the partial least squares calibration error. Furthermore, the variation in the signal-to-noise ratio of spectra sensitively influences the calibration. For reliable prediction, it is critical to maintain the signal-to-noise ratio at a certain level. When the sample spectra were collected at a higher temperature, it was possible to acquire spectra with an improved signal-to-noise ratio and better calibration. In addition, a single beam spectrum of water shifts to a lower frequency, and the maximum transmission intensity at around 2500 cm^{-1} (the heavy water band location) increases at an elevated temperature. Overall, an on-line infrared spectroscopic scheme is presented for measuring heavy water. The scheme can be applied to an actual process without practical difficulties. If the spectra could be collected at elevated temperature over 2 min with the use of a high throughput light source, the prediction error could reach to 1.0 ppm.

Introduction

In an earlier publication, we demonstrated the use of infrared (IR) spectroscopy to measure heavy water in a nuclear power plant.[1] With the aid of partial least squares (PLS) regression,[2–5] we achieved an error of 3.3 ppm in the 150–250 ppm range. In the previous study, which was based on simple laboratory feasibility, we did not consider other important parameters that could influence the accuracy of the measurement. The goal of this paper, therefore, is to study and identify the important spectroscopic parameters for on-line measurement, such as temperature and signal-to-noise variations. We hope to ultimately improve the accuracy of IR measurement for practical industrial application.

Initially, we systematically varied the signal-to-noise ratios of spectra by changing the number of scans. We also collected the sample spectra at room temperature by using a flow cell to simulate on-line analysis. Our aim was to examine the relationship between the resulting calibration error and the corresponding signal-to-noise ratio. During this study, we observed that temperature variation was another critical parameter that should be controlled tightly because the shift in the water band from a minor temperature change causes non-linear baseline variations that influence the calibration. We therefore built an improved on-line flow cell system with tighter temperature control, and we collected the corresponding spectra. The resulting calibration error was improved by minimizing the baseline variations from even minor temperature changes.

Finally, we found that the absolute temperature of measurement influences the signal-to-noise ratio of a spectrum because of the optical transmission window, which is located between the big water bands. This window, which enables us to measure heavy water, is increased due to the shift in the water band.[6–8] At an elevated temperature, the shift in the water band provides more light throughput in the spectral range for heavy water analysis (that is, the 2700–2400 cm^{-1} range). As a result, a higher signal-to-noise ratio improves the calibration.

Experimental

Sample preparation

We prepared 37 samples with a concentration ranging from 130 ppm to 180 ppm. We purchased D_2O of 99.9 atom percent from Sigma–Aldrich, and mixed an appropriate amount of it with distilled H_2O. To minimize analytical errors in the sample preparation, we used a 1 L volumetric flask to prepare all the samples on a large scale. Throughout the entire experiment, we used ultra-pure water (>18 MΩ cm^{-1}, Milli-Q).

Spectral collection and data processing

To collect FT-IR spectra, we used an ABB Bomem 100 spectrometer (ABB Bomem Inc., Quebec City, Quebec, Canada) equipped with a globar source and DTGS detector. Each spectrum was collected with the aid of a transmission cell made of CaF_2 and a path length of 0.2 mm, as determined elsewhere. The spectral resolution was fixed at 8 cm^{-1} with a point spacing of 3.86 cm^{-1}.

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For the PLS regression, we used GRAMS/32 software with an add-on PLS algorithm (Galactic Industries Corporation, Salem, NH, USA). We imported the IR spectra into GRAMS/32 before the PLS regression. Of the 74 spectra (two for each sample) we used 54 for calibration and 20 for the set of prediction data to serve as a validation set.

Results and discussion

Spectral features

When IR spectroscopy is used for on-line monitoring of heavy water in real nuclear power plants, the measurements are performed on a flowing sample to minimize the delay between the process conditions and the corresponding analytical measurement. As a result, we initially examined the spectral differences between when the sample was stagnant and flowing. From this examination, we expected no significant differences in the general spectral features, except for the possibility of some minor changes such as signal-to-noise variation. Fig. 1 shows IR spectra (32 scans) of the 150 ppm heavy water sample under stagnant and flowing conditions (with a flow rate of 20 ml min^{-1}). Both spectra are similar to each other, although the spectrum acquired with the flowing sample is just slightly noisier because of the small light attenuation, which was probably due to refraction and scattering.

The root mean square (RMS) noise was calculated for a quantitative comparison.[9] For this purpose, we acquired 30 zero absorption lines (baselines) by ratioing two consecutive water single beam spectra for stagnant and flowing samples with three different flow rates (3 ml min^{-1}, 6 ml min^{-1} and 20 ml min^{-1}). To calculate the RMS noise, we then divided the 2700 cm^{-1} to 2400 cm^{-1} range (the range used for heavy water analysis) into five regions (2700 cm^{-1} to 2640 cm^{-1}, 2640 cm^{-1} to 2580 cm^{-1}, 2580 cm^{-1} to 2520 cm^{-1}, 2520 cm^{-1} to 2460 cm^{-1} and 2460 cm^{-1} to 2400 cm^{-1}) because the corresponding baselines might not have been straight in some cases. In the five regions, the baselines were straight and we successfully used the least square method to calculate the RMS noise. The overall RMS noise for the 2700 cm^{-1} to 2400 cm^{-1} range was the average of the RMS noise for the five regions.

To calculate the final RMS noise, we averaged the 30 RMS noise values. The calculated RMS noises at 0, 3, 6 and 20 ml min^{-1} were 0.000 16, 0.000 18, 0.000 19 and 0.000 18, respectively. As expected from Fig. 1, the RMS noise was slightly lower for the stagnant sample. For the flowing samples, the resulting RMS noises were slightly increased, though no critical difference was observed when the flow rate was changed. We collected spectra of samples flowing at 20 ml min^{-1} for practical purposes, even though a slight degradation occurred in the signal-to-noise ratio.

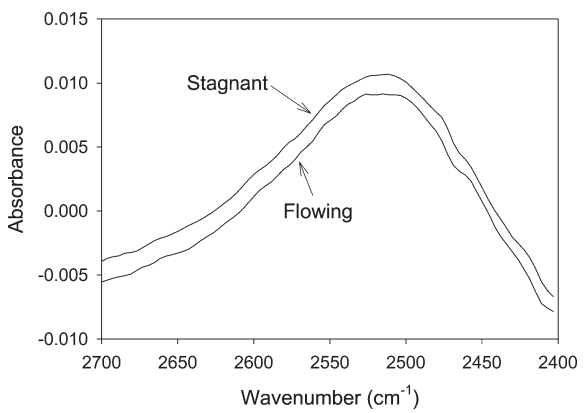

Fig. 1 IR spectra (32 scans) of the 150 ppm heavy water sample under stagnant and flowing conditions (20 ml min^{-1}).

Temperature variation

The slight temperature variation between the background and the sample single beam spectra collection can introduce the baseline variation, especially at the right part (2500 cm^{-1} to 2400 cm^{-1}) of the absorption band, because of the overlap with the water band, which shifts with variations in the temperature. Because the sample is mostly composed of water, even a minor band shift can easily affect the baseline of the heavy water band, which has significantly low absorption.

Fig. 2(a) shows the water single beam spectra collected at 24 °C, 25 °C, 27 °C and 29 °C. Clearly, the water band shifts to the lower frequency even under small variations of temperature. The absolute magnitude of the shift might not be significant but could greatly impact on the heavy water spectrum. Fig. 2(b) shows the baselines (zero absorption lines) at various temperatures ratioed with a single beam spectrum at 24 °C. We observed apparent nonlinear baseline variations in the 2700 cm^{-1} to 2400 cm^{-1} range, and the degree of variation

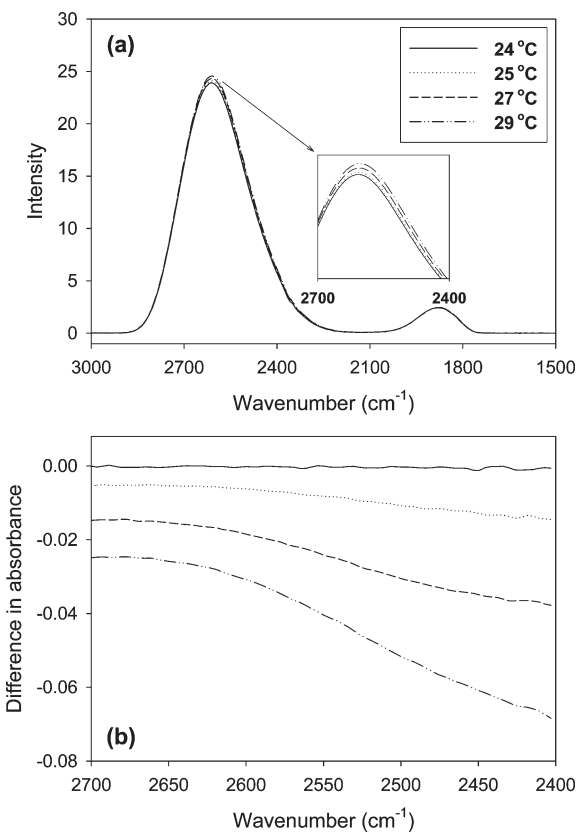

Fig. 2 Water single beam spectra collected at 24, 25, 27 and 29 °C (a), and baselines (zero absorption lines) at various temperatures ratioed with a single beam spectrum at 24 °C (b).

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increases as the temperature difference increases. When the single beam spectra at various temperatures are ratioed to the single beam spectrum of the highest temperature (29 °C), the trend is exactly reversed. When the temperature is changed, the final spectrum is a summation of pure heavy water absorption and the underlying baseline variation. For reliable prediction, the baseline variation from the temperature change should be corrected mathematically or a tight system of temperature control is required. As will be discussed later, the position of the heavy water band was insensitive to temperature variation.

On-line measurement system

To simulate an actual process, we initially used a syringe pump to enable the sample to flow into a cell without a temperature control unit. All the measurements were taken at room (with an air-conditioning system) temperature, and the sample temperature was measured, with the aid of a thermocouple, at the outlet of the flow cell. As a result, we controlled the temperature within a range of approximately ± 2.0 °C (refer to Experiment A). To improve the capability of the temperature control, we built an on-line system of measurement. Fig. 3 shows the corresponding schematic diagram (refer to Experiment B). We generated the sample flow by using pressurized air that was connected inside tightly sealed reference and sample containers (with a volume of 3 L). To control the flow rate, we used a regulator (20 ml min^{-1}) and, to maintain the temperature of the sample at a constant by heat exchange, we used a 1 m coil of stainless steel tubing inside a water bath. With a constant temperature, the samples were fed into an IR transmission flow cell. The temperature at the outlet of the IR cell was measured with the aid of a thermocouple and recorded periodically. By using the system, we controlled the temperature with a range of ± 0.5 °C.

To collect the IR spectra, we used the schemes described in Experiment A and Experiment B. The resulting IR spectra (32 scans) were compared. Fig. 4 shows 40 randomly selected

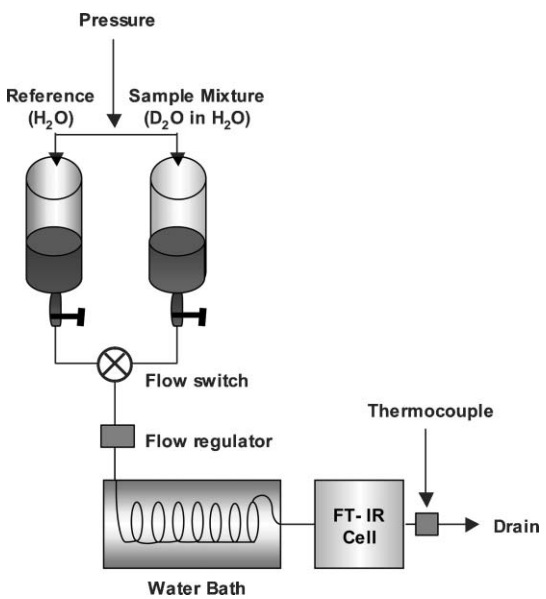

Fig. 3 Schematic diagram for on-line measurement of water (Experiment B).

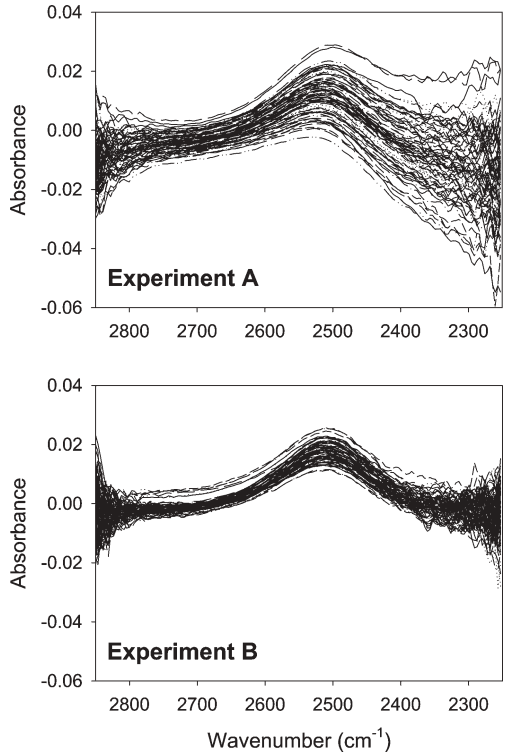

Fig. 4 Forty randomly selected spectra collected from both Experiment A and B.

spectra (with the same absorbance scale) from both experimental schemes. The baseline variations are clearly larger for Experiment A. Although it was difficult to quantitatively relate the calibration error and the degree of temperature variation, we had to minimize the baseline variations.

PLS regression was used to build the calibration model and to compare the results of Experiments A and B. For the spectral range, we used only the 2700 cm^{-1} to 2400 cm^{-1} range, which included the only heavy water information. For the optimum number of factors, we chose the value that gave the minimum standard error of cross validation (SECV). We applied the cross validation method by dividing the data set into five segments, and we examined the pattern of the decreasing SECV as a function of the number of PLS factors. The SECV decreased sharply for the initial factors and then decreased gradually for the following factors. At a certain factor, the value of the SECV began to increase. We chose the factor before the increase in the value of the SECV as the optimum number of factors.

The resulting SECVs from Experiment A and Experiment B were 2.7 ppm and 2.0 ppm, respectively. Three factors were used for both cases to achieve the optimal calibration. The SECV from Experiment B was improved when the temperature of the sample was more tightly controlled. Based on these results, the temperature is an important factor that governs the performance of the calibration and it should be controlled as strictly as possible.

Influence of signal-to-noise variation on calibration performance

The signal-to-noise ratio is another critical factor that determines the calibration performance, especially in this study,

because the magnitude of the absorption and the increase of absorption that results from variations in concentration should be small for measurements of low concentration (in the ppm range). In a given instrumental setting, the signal-to-noise ratio can be improved by increasing the number of scans; however, this improvement is related to the analysis time when applied to practical on-line measurements.

To study the relationship between the signal-to-noise ratio and the calibration error, we varied the number of scans from 16 to 32 to 64, and we collected the corresponding IR spectra by using the scheme of Experiment B. The 16, 32 and 64 scans corresponded to 30 s, 60 s and 120 s for the spectral collection.

The signal-to-noise ratio is theoretically proportional to the square root of the number of scans; however, the improvement of signal-to-noise ratio cannot be compared to the theoretical value in reality due to the drift effect. We have calculated the RMS noise for quantitative comparison at each scan number and the corresponding calibration results are summarized in Table 1, which shows that the RMS noise decreases as the number of scans increases. The numbers in parenthesis correspond to the number of factors used. In this case, we used the calibration set to estimate the SECV and we calculated the SEP by predicting samples in the prediction set. As the RMS noise decreased, the resulting SEP clearly improved. If we allow 2 min for the analysis (64 scans) with the existing system, we can achieve a prediction error of 1.9 ppm for the on-line measurement. When we apply this methodology to a real nuclear power plant, we must improve and maintain the signal-to-noise ratio at a certain level for reliable and robust measurement.

Measurement at an elevated temperature

All the on-line studies were accomplished at around room temperature (24 °C). Fortunately, on the basis of Fig. 2, we can increase the optical throughput near the heavy water band (2500 cm^{-1}) by simply increasing the temperature at which we take the measurements. The single beam spectrum of water shifts to the lower frequency, and the maximum transmission intensity is increased as the temperature increases. This allows more light throughput at around 2500 cm^{-1} and eventually improves the signal-to-noise ratio of the resulting spectrum.

Fig. 5 shows single beam and absorbance spectra of 150 ppm heavy water at 24 °C and 43 °C. The absorbance spectra (32 scans) were collected without a flowing sample. The temperature of the sample was more tightly controlled ($\pm$0.2 °C) with the aid of a new IR cell surrounded with a water jacket. We could not measure the spectra above 43 °C because of the presence of air bubbles when we collected the spectra. The air bubbles caused a severe baseline offset from the scattering. In our configuration, 43 °C was the highest temperature at which we could obtain spectra without air bubbles. By increasing the temperature, we greatly increased the optical throughput and acquired spectra that had a better signal-to-noise ratio. Furthermore, the position of the heavy water band was insensitive to temperature variation.

Table 1 RMS noises and corresponding calibration results based on different numbers of scans

Number of scans	RMS noise	SECV (ppm)	SEP (ppm)
16	0.000 24	2.2 (4)	3.0
32	0.000 17	2.0 (4)	2.2
64	0.000 12	1.4 (4)	1.9

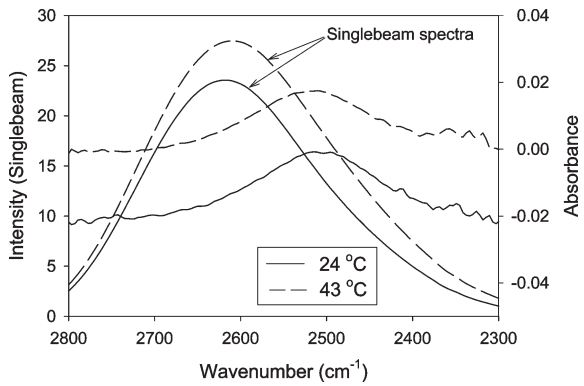

Fig. 5 Single beam and absorbance spectra of 150 ppm heavy water at 24 and 43 °C.

Table 2 The calculated RMS noises at two different temperatures and PLS calibration results for spectra collected at 24 °C and 43 °C

Temperature/°C	RMS noise	SECV (ppm)	SEP (ppm)
24	0.000 18	1.6 (3)	1.9
43	0.000 14	1.3 (3)	1.6

Table 2 shows the calculated RMS noise at two different temperatures. For easier comparison, these calculations are based on the spectra that were taken without a flowing sample. Clearly, the RMS noises decreased at a higher temperature. We expected no significant spectral differences between the stagnant and flowing sample, as shown in Fig. 1.

For the PLS we used the spectra (32 scans) collected at 24 °C and 43 °C. At each temperature, we collected 74 spectra from 37 samples in the manner described in Table 1 (54 spectra for calibration and 20 spectra for validation). Table 2 shows the results. The SEP at 43 °C was clearly improved due to the improved signal-to-noise ratio. The SEP at 24 °C without a flowing sample (32 scans) is better than the SEP taken under the same conditions as Experiment B (Table 1), and even the RMS noises are similar to each other. This result, which was derived from the tighter temperature control of the stagnant sample, led to a more stable baseline. If we can build a pressured sampling system or some other technique that can suppress the generation of bubbles at a higher temperature than 43 °C, the SEP will be improved further.

Conclusion

For the measurement of heavy water, we studied the important parameters of variations in temperature and in the signal-to-noise ratio. As a result, we improved the prediction accuracy for a SEP of 1.6 ppm by measuring samples at 43 °C during a collection period of 1 min (32 scans). Strict temperature control is critical. Furthermore, because we used a clear liquid for the process stream, the application is feasible in an actual process stream. In practice, it would not be difficult to design an on-line sampling system with strict temperature control and

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bubble suppression (or removal) at an elevated temperature. If we can implement dual IR light sources (to increase the radiation intensity) and collect spectra over a period of more than 2 min (64 scans) at a temperature above 43 °C with strict temperature control, we can eventually achieve a SEP below 1.0 ppm.

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